

4 Discrete Memoryless Channel and Noisy Channel Coding Theorem

Reading: Chapter 7.1–7.10 of Cover and Thomas

Preview of Lec. 15–18

- Discrete Memoryless Channel (DMC)
- Joint AEP
- Channel coding theorem

The main difference between this part of the course and the previous ones is that we are now dealing with two (sequences of) random variables instead of one. For example, we consider the inputs and outputs of a channel. Thus, our main focus is on the relation between two sequences, and to describe it using the joint distribution.

4.1 Discrete Memoryless Channel (DMC) and Channel Capacity

Definition 1 (Discrete Memoryless Channel (DMC)). A channel is the (physical or abstract) link connecting input $X \in \mathcal{X}$ and output $Y \in \mathcal{Y}$, described by the conditional probability $W(y|x)$ with output y for input x . A discrete channel takes one input symbol X_i and yields one output symbol Y_i at each time, when $i = 1, 2, \dots$ is the time index. The channel is said to be *memoryless* if the probability distribution of the output depends only on the input at the current time and independent of previous channel inputs or outputs. A Discrete Memoryless Channel (DMC) satisfies

$$\Pr(y^n|x^n) = \prod_{i=1}^n W(y_i|x_i), \quad (1)$$

and is written as $DMC(W)$.

Remarks:

- 1) There is a notion of continuous time channel, which takes continuous time waveforms as input and output. By Nyquist–Shannon sampling theorem, we can convert the continuous time channel to discrete time channel by sampling the waveforms.
- 2) The input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$ are in most cases assumed to be finite, except when we talk about Gaussian channels. In the case of continuous random variables, W can be thought of as a conditional density of Y given X . For discrete input and output, W can be thought of as a stochastic matrix, where each row is a conditional distribution of Y , conditioned on a particular $x \in \mathcal{X}$. We sometimes write a row vector of W as W_x .
- 3) Notations: We write $P_X \cdot W$ as the joint distributions of the input and the output and $P_X \cdot P_Y$ as the product of marginals. For n uses of the channel W , the induced conditional distribution between the input sequence x^n and output sequence y^n is written as $W^n(y^n|x^n)$.
- 4) Examples of channels can be quite general. Besides common communication channels for wireless/optical communications and internet packet transmissions, other problems such as data storage, data collection/information extraction can also be viewed as a channel.

Definition 2 (Channel Capacity). For a given $DMC(W)$, the channel capacity is defined as

$$C \triangleq \max_{P_X} I(P_X, W) \text{ bits/channel use.} \quad (2)$$

Remarks:

- 1) **Important remark:** At this point, the capacity of a channel is nothing but a quantity associated with the given channel, with a nice name. We will show later that this quantity has the operational meaning that a information flow with rate $R < C$ can be communicated over the channel “reliably”, and a stream with $R > C$ cannot. Before proving this, although we call the quantity capacity, it doesn’t mean anything.
- 2) Recall that $I(P_X, W)$ is concave in P_X for a fixed W . Thus, the optimization has a nice form.
- 3) Optimal input distribution: $P_X^* = \arg \max_{P_X} I(P_X, W)$.
- 4) Numerical solution to the optimization problem exists for channels with finite input and output alphabets, which is called Blahut-Arimoto algorithm. However, the precise numerical evaluation of the capacity as a number has little use, since capacity itself only has the operational meaning in asymptotic regions. Instead, the importance of capacity often lies in the insights one can obtain from understanding what would be the optimal code and how it should look like.

Before going to any example, one should ask the following question: For a given DMC, one can always induce a more complicated channel by using k symbols at a time. For the new channel with input and output alphabets $\mathcal{X}^k$ and $\mathcal{Y}^k$ and the conditional probability matrix W^k , one can easily define

$$C^{(k)} \triangleq \frac{1}{k} \max_{P_{X^k}} I(P_{X^k}, W^k) \quad (3)$$

where the factor $1/k$ is used to make a fair comparison to the single symbol case. The question is whether $C^{(k)}$ can be larger than C defined for a single symbol. If we think the capacity C as the amount of bits one can maximally communicate over the channel, it only makes sense if $C^{(k)} = C$. If $C^{(k)} > C$ in any case, then C cannot be a good definition for channel capacity. Thus, we consider

$$\begin{aligned} I(X^k; Y^k) &= H(Y^k) - H(Y^k | X^k) \\ &= H(Y^k) - \sum_{i=1}^k H(Y_i | X^k, Y_1^{i-1}) \\ &= H(Y^k) - \sum_{i=1}^k H(Y_i | X_i) \quad \text{by the memoryless of the channel} \\ &\stackrel{\substack{\text{iff } Y_i \text{'s are indep.} \\ \text{or } X_i \text{'s are indep.}}}{\leq} \sum_{i=1}^k (H(Y_i) - H(Y_i | X_i)) \quad \text{conditioning only reduces the entropy} \\ &= \sum_{i=1}^k I(X_i; Y_i) \\ &\stackrel{\text{all } P_{X_i} = P_X^*}{\leq} kC. \end{aligned} \quad (4)$$

The first $\leq$ takes equality iff Y_i ’s are independent, which can be obtained if X_i ’s are independent. The second $\leq$ takes equality iff P_{X_i} are all equal to the optimal P_X^* . This proves that $C^{(k)}$ equals C , and the optimal k symbol input distribution is i.i.d. with the single symbol optimal input distribution. This result heavily depends on the memoryless structure of the channel, and it is not true for channels with memory.

Example: [Binary Symmetric Channel (BSC)] We denote by $\text{BSC}(\epsilon)$ the following channel: $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and

$$W(y|x) = \begin{cases} 1 - \epsilon & \text{if } x = y \\ \epsilon & \text{otherwise.} \end{cases} \quad (5)$$

One can write this channel differently as

$$Y_i = X_i \oplus Z_i \quad (6)$$

where $\oplus$ is XOR and Z_i 's are i.i.d. binary random variable with Bernoulli(ϵ) distribution, independent of X_i 's. In words, the channel flips the input bit with probability ϵ .

To find the capacity of this channel, we write

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - H_B(\epsilon) \\ &\leq \log 2 - H_B(\epsilon) \end{aligned} \quad (7)$$

where the equality is achieved iff $P_Y = [1/2, 1/2]$. This can be obtained by choosing $P_X = [1/2, 1/2]$. Thus, the capacity is

$$C = \log 2 - H_B(\epsilon) \quad (8)$$

and the optimal input distribution is Bernoulli(1/2). [PICTURE: Capacity with $\epsilon \in [0, 1]$.]

Remarks:

- 1) The worst channel is BSC(1/2), with capacity 0, and the best is BSC(0), with capacity 1 bit/channel use.
- 2) The optimal input distribution of BSC(ϵ) can be proven by using symmetry of the channel. Note that $P_X = [1 - p, p]$ and $P_X = [p, 1 - p]$ result in the same mutual information; then since $I(P_X, W)$ is concave in p , the maximizer should be $[1/2, 1/2]$.

Example: [Binary Erasure Channel (BEC)] Consider the channel with $\mathcal{X} = \{0, 1\}$ and $\mathcal{Y} = \{0, 1, E\}$ where E is called the erasure symbol. BEC(ϵ) is defined as

$$W(y|x) = \begin{cases} 1 - \epsilon & \text{if } x = y \\ \epsilon & \text{if } y = E. \end{cases} \quad (9)$$

In words, the channel never flip an input bit, but with a probability ϵ it yields an erasure (not-sure) output. For this channel,

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= H(X) - [\Pr(Y = E)H(X) + \Pr(Y \neq E)H_B(0)] \\ &= H(X) - \epsilon \log 2. \end{aligned} \quad (10)$$

Now we can simply maximize $H(X)$ with $P_X = [1/2, 1/2]$. Compare the capacity of BSC(ϵ) and BEC(ϵ) for $0 < \epsilon \leq 1/2$. We can see that the capacity of BEC is always better. This is intuitive since for BEC we know when we are not sure of the transmitted input, which certainly makes reliable communication easier.

* * *

Discussions: We argued that capacity has the operational meaning of the maximum rate (bits/channel use) that can be communicated reliably over the channel. Let's now make clear what we mean by "reliable" communications. Consider transmitting a single bit over a BSC(ϵ) with $\epsilon < 1/2$. A receiver makes a guess given the output of the channel. Clearly, the receiver has an error probability ϵ in guessing the transmitted input symbol. Now one can repeatedly send the same bit n times by n uses of the channel. As a cost, the data rate is $R = 1/n$ (bits/channel

use). The receiver guesses that 0 was transmitted if the received bits has more than $n/2$ 0's. Otherwise, the receiver guesses that 1 was transmitted. For this case the error probability is

$$P_e^{(n)} = \sum_{k=n/2}^n \binom{n}{k} (1-\epsilon)^{n-k} \epsilon^k$$

$$\approx e^{n \log 2 (1-\epsilon)^{n/2} \epsilon^{n/2}} = e^{-n(-\log 2 \sqrt{\epsilon(1-\epsilon)})}. \quad (11)$$

This was pretty much the story before Shannon, that there is a natural trade-off between the rate and reliability (error probability).

On the other hand, Shannon observed that arbitrary small $P_e^{(n)}$ can be achieved without necessarily reducing the data rate *as the codeword length $n \rightarrow \infty$* . He showed that with rate $R < C$ one can find a communication scheme (channel codes) that can make $P_e^{(n)} = e^{-n\delta}$ for some $\delta > 0$. Therefore, as $n \rightarrow \infty$, $P_e^{(n)} \rightarrow 0$. So, Shannon defined “reliable communication” as that the probability of error goes to 0 as the codeword length goes to infinity. Moreover, Shannon also showed that such reliable communication can never happen when the rate $R > C$. Shannon’s contribution is on brining the one more knob (n , the number of channel uses) into the picture and deriving meaningful conclusions in the extreme case where $n \rightarrow \infty$. ← (reliable communication)

Is there really no tradeoff between the data rate and reliability? Given Shannon’s result, it seems like there is no tradeoff as long as $R < C$. This is not really true. Gallager later defined the notion of error exponent, which indicates how fast the error $P_e = e^{-nE(R)}$ decays given rate $R < C$, where $E(R)$ is a positive exponent and called error exponent, which is a function of the rate R . Thus, there exists trade-offs between R , $P_e^{(n)}$, and n in the communication problem.

4.2 Weak Joint Typicality

Definition 3 (Weak Joint Typical Set). For a given joint distribution P_{XY} on $\mathcal{X} \times \mathcal{Y}$, let P_X and P_Y be its X and Y marginals, respectively. The weak joint typical set $A_\epsilon^{(n)}(P_{XY})$ is defined as

$$A_\epsilon^{(n)}(P_{XY}) \triangleq \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \left| -\frac{1}{n} \log P_{XY}^n(x^n, y^n) - H(P_{XY}) \right| \leq \epsilon, \quad (12)$$

$$\left| -\frac{1}{n} \log P_X^n(x^n) - H(P_X) \right| \leq \epsilon, \quad (13)$$

$$\left| -\frac{1}{n} \log P_Y^n(y^n) - H(P_Y) \right| \leq \epsilon \} \quad (14)$$

Remarks: Compare the definition for the weak joint typical set to the definition for weak typical set $A_\epsilon^{(n)}(P_X)$ for one random variable X . Can you see something strange? When we define some mathematical quantity, in general, there should be no difference between the definition for a random variable and the definition for a random vector. Thus, a natural definition for a joint typical set should include only the condition like (12), regarding the joint probability of the pair of sequences. This is true if we have strong typicality, which will be discussed later. Here, (13) and (14) are needed since they are not implied from (12). We will continue with this definition to prove channel coding theorem in the way Cover’s book does. One should keep an eye on why all these conditions are needed. We will talk about the strong AEP in the next section, which is a good comparison.

Theorem 1 (Joint AEP). Three facts about $A_\epsilon^{(n)}(P_{XY})$

1. $P_{XY}^n(A_\epsilon^{(n)}(P_{XY})) \rightarrow 1$ as $n \rightarrow \infty$

2. $|A_\epsilon^{(n)}(P_{XY})| \leq e^{n(H(P_{XY})+\epsilon)}$;
 2.5 For large enough n , $|A_\epsilon^{(n)}(P_{XY})| \geq (1-\epsilon)e^{n(H(P_{XY})-\epsilon)}$
3. $[P_X \cdot P_Y]^n(A_\epsilon^{(n)}(P_{XY})) \doteq e^{-nI(X;Y)}$

Proof. The proofs of property 1 and 2 are exactly parallel to their counterparts for the single random variable case. The only thing one should keep in mind is that if $(X_i, Y_i)_{i=1}^n$ are i.i.d. and drawn from the joint distribution P_{XY} , then X_i 's are i.i.d. from P_X and Y_i 's are i.i.d. from P_Y , thus the extra constraints (13) and (14) does not change (reduce) the typical set much.

The only thing interesting is statement 3. Recall that the notation means that evaluated with distribution $[P_X \cdot P_Y]$ (the product of two marginals), the joint typical set with respect to (w.r.t) P_{XY} has a certainly small probability. This statement is trivial if we have strong AEP and apply Sanov's theorem: the probability that a length- n sequence drawn from a $[P_X \cdot P_Y]$ distribution appears typical w.r.t. P_{XY} is simply $e^{-nD(P_{XY}||[P_X \cdot P_Y])} = e^{-nI(X;Y)}$. Now for the weak AEP definition, we need to write

$$\begin{aligned}
 [P_X \cdot P_Y]^n(A_\epsilon^{(n)}(P_{XY})) &= \sum_{(x^n, y^n) \in A_\epsilon^{(n)}(P_{XY})} P_X^n(x^n) P_Y^n(y^n) \\
 &\doteq |A_\epsilon^{(n)}(P_{XY})| e^{-nH(P_X)} e^{-nH(P_Y)} \\
 &\doteq e^{-n(H(P_X)+H(P_Y)-H(P_{XY}))} \\
 &= e^{-nI(X;Y)}
 \end{aligned} \tag{15}$$

where the second line follows from (13) and (14), and the third line follows from the property 2 and 2.5 in the theorem. $\square$

4.3 Formal Definitions of Channel Coding and Channel Coding Theorem

Definition 4. For a given DMC(W), with input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$, an (n, R) channel code is a pair of maps, f_n (encoder) and g_n (decoder) where

$$\begin{aligned}
 f_n &: \{1, 2, \dots, 2^{nR}\} \rightarrow \mathcal{X}^n \\
 g_n &: \mathcal{Y}^n \rightarrow \{1, 2, \dots, 2^{nR}\}.
 \end{aligned}$$

Remarks:

- 1) For message $m \in \{1, 2, \dots, 2^{nR}\}$, $f_n(m)$ is called the m -th codeword. We often write $x^n(m) \in \mathcal{X}^n$ for the m -th codeword. The collection of codewords are often written as a matrix of size $2^{nR} \times n$, which is called a codebook.
- 2) The actual communication happens in the following way:
 - The codebook, or equivalently, f_n , is designed ahead of time. This is known both to the encoder and the decoder.
 - When communication starts, a random variable M , uniformly distributed in $\{1, 2, \dots, 2^{nR}\}$, or equivalently, nR bits, are revealed to the encoder. The possible values of M are called messages. The encoder maps it to a codeword $x^n(m)$, and transmit it by n uses of the channel.
 - The receiver observes the channel output y^n and apply the decoding g_n to make a guess of the value of the message $\hat{M} = g_n(y^n)$.
 - We say an error occurs if $\hat{M} \neq M$. The goal of the design of channel codes is to send the data with a high rate, and a small probability of error.

- 3) For given f_n , the optimal decoding rule g_n is clear. It is well known that the maximum likelihood decoder minimizes the error probability (note that all messages are equally likely):

$$g_{\text{ML}}(y^n) = \arg \max_m W^n(y^n | x^n(m)). \quad (16)$$

It turns out that this decoder is a bit hard to analyze, so we often use sub-optimal decoders, which are easier to analyze but not too much worse.

Definition 5. For a given (n, R) code, the conditional error probability, conditioned on a particular message $m \in \{1, \dots, 2^{nR}\}$, is

$$P_{e,m}^{(n)} \triangleq \Pr(g_n(y^n) \neq m | x^n(m)), \quad (17)$$

and the maximum error probability is

$$P_{e,\max}^{(n)} \triangleq \max_m P_{e,m}^{(n)}. \quad (18)$$

The average error probability is

$$P_e^{(n)} = \frac{1}{2^{nR}} \sum_{m=1}^{2^{nR}} P_{e,m}^{(n)}. \quad (19)$$

Definition 6. We say a rate R is achievable if there exists a sequence of (n, R) codes such that

$$P_{e,\max}^{(n)} \rightarrow 0 \quad (20)$$

as $n \rightarrow \infty$.

Remarks:

- 1) The notion of a sequence of codes
- 2) It is important to note that we insist the maximum probability of error, over all possible messages, to decay to 0, thus ensure a uniform reliability over all messages. The way to prove that a coding scheme achieves this, however, often starts from showing the average error probability $P_e^{(n)} \rightarrow 0$.

Theorem 2 (Channel Coding Theorem). The maximum achievable rate is $C = \max_{P_X} I(P_X, W)$. To be precise, if $R < C$, then R is achievable, and if $R > C$, then it is not achievable.

Proof [Converse:] The proof of the converse is actually very simple, using Fano's inequality. Let M denote the message fed to the encoder. We have

$$\begin{aligned} nR = H(M) &= H(M|Y^n) + I(M; Y^n) \\ &\leq H(M|Y^n) + I(X^n(M); Y^n) \quad \text{data processing inequality} \\ &\leq \log 2 + P_e^{(n)} nR + I(X^n(M); Y^n) \quad \text{Fano's inequality} \\ &\leq \log 2 + P_e^{(n)} nR + nC \quad \text{By the definition of } C \text{ and the memoryless property} \end{aligned} \quad (21)$$

where $P_e^{(n)}$ is the average error probability of guessing M based on the observation of Y^n . By rearranging the terms, we get

$$P_e^{(n)} \geq 1 - \frac{C}{R} - \frac{1}{nR}. \quad (22)$$

If $R < C$, this inequality is useless; if $R > C$, the right hand side is bounded away from 0.

Remarks:

- 1) This converse only says that when $R > C$ one cannot push the error probability to 0 if n is getting large. Can we push it to zero at some finite n ? If we have a good code like that at any finite n , it is not hard to use the code to create good codes of length $2n, 3n, \dots$. With a little thinking, it is not hard to verify that the above proof also rules out the possibility of reliable communication at rate above the capacity at any finite n .
- 2) The converse states that at rate above the capacity, one cannot push the average error probability to zero. This is a stronger statement than needed, and it clearly implies that one cannot push the maximum error probability to zero.

Proof [Achievability:]

Idea: Random Coding

First of all, one has to realize that it is very hard to construct a codebook of such a high dimension $2^{nR} \times n$, for which we'd like to guarantee $P_{e,\max}^{(n)} \rightarrow 0$. To some extent such objects can be handled only if there is some kind of nice symmetry. To create such a symmetry, random coding is used.

So the idea is to randomly generate the codebook according to some distribution. Thus we would have an ensemble of codebooks. The probability of having a particular codebook $\mathcal{C}$ is written as $\Pr(\mathcal{C})$. If we could show that average error probability over this ensemble satisfies

$$\sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr(\text{error}|\mathcal{C}) \leq \epsilon, \quad (23)$$

this implies that at least one of these codebooks, say $\mathcal{C}^*$, would have $\Pr(\text{error}|\mathcal{C}^*) \leq \epsilon$.

Code Construction:

Fix a distribution P_X and generate a codebook $\mathcal{C}$ as a $2^{nR} \times n$ matrix, with i.i.d. entries from P_X . The codebook is fixed once generated and revealed to the decoder.

Transmission:

As defined before, when transmitting a message $m \in \{1, \dots, 2^{nR}\}$, we pick the m -th row of the codebook, $x^n(m) \in \mathcal{X}^n$, and send it over n uses of the channel.

Decoding:

As discussed before, the optimal decoding is the Maximum Likelihood (ML) decoding. However, for the sake of analysis, we use a sub-optimal decoding: joint typicality decoding. Upon receiving the channel output y^n , the receiver checks for each possible value of m , whether $x^n(m)$ is jointly typical with y^n , w.r.t., $P_X \cdot W$, i.e., whether $(x^n(m), y^n) \in A_\epsilon^{(n)}(P_{XY})$.

- If there exists a unique m such that $(x^n(m), y^n) \in A_\epsilon^{(n)}(P_{XY})$, the decoder claims that such a message m is transmitted;
- If there does not exist such a m , or if there exist more than one codeword jointly typical with y^n , the receiver declares an error.

Remarks:

- 1) Note that this decoding rule is very strict: if any codeword $x^n(m)$ is generated to be atypical, i.e., $x^n(m) \notin A_\epsilon^{(n)}(P_X)$, then $P_{e,m}^{(n)} = 1$ and thus $P_{e,\max}^{(n)} = 1$. The only good thing is that there is only a small probability that a codeword $x^n(m)$ is atypical.
- 2) If y^n is atypical to x^n for some reason, there will always be an error. For example, if the channel behaves so good that a BSC(ϵ) does not flip any bit at all, it would cause an error.

Error Probability Analysis:

When message m is transmitted, an error occurs iff either of the following events occurs:

$$\begin{aligned}\mathcal{E}_1 &\triangleq \{(x^n(m), y^n) \notin A_\epsilon^{(n)}(P_{XY})\}, \\ \mathcal{E}_2 &\triangleq \{\exists m' \neq m, \text{ s.t., } (x^n(m'), y^n) \in A_\epsilon^{(n)}(P_{XY})\}.\end{aligned}\tag{24}$$

Now we have

$$\begin{aligned}\Pr(\text{error}|m, \mathcal{C}) &= \Pr(\mathcal{E}_1 \cup \mathcal{E}_2|m, \mathcal{C}) \\ &\leq \Pr(\mathcal{E}_1|m, \mathcal{C}) + \Pr(\mathcal{E}_2|m, \mathcal{C}). \quad \leftarrow \text{union bound.}\end{aligned}\tag{25}$$

Now here we use symmetry: we compute the average probability of error, averaged over all m and all $\mathcal{C}$,

$$\begin{aligned}\Pr(\text{error}) &= \sum_{\mathcal{C}} \Pr(C) \sum_{m=1}^{2^{nR}} \Pr(m) \Pr(\text{error}|m, \mathcal{C}) \\ &= \frac{1}{2^{nR}} \sum_{m=1}^{2^{nR}} \underbrace{\sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr(\text{error}|m, \mathcal{C})}_{\text{doesn't depend on } m} \\ &= \sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr(\text{error}|m=1, \mathcal{C}) \quad \text{w.l.o.g (without loss of generality)} \\ &\leq \sum_{\mathcal{C}} \Pr(\mathcal{C}) [\Pr(\mathcal{E}_1|m=1, \mathcal{C}) + \Pr(\mathcal{E}_2|m=1, \mathcal{C})].\end{aligned}\tag{26}$$

Note that

$$\sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr(\mathcal{E}_1|m=1, \mathcal{C}) = \Pr[(X^n(1), Y^n) \notin A_\epsilon^{(n)}(P_{XY})|m=1] \rightarrow 0.\tag{27}$$

The trick here is that when averaged over $\mathcal{C}$, the codeword $X^n(1)$ becomes random with a simple i.i.d. P_X distribution. The channel output Y^n is then i.i.d. by $W^n(\cdot|X^n(1))$. By this trick, we avoid talking about a specific length- n string, but rather work on single letter i.i.d. distribution.

Now consider the second error type.

$$\begin{aligned}&\sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr(\mathcal{E}_2|m=1, \mathcal{C}) \\ &\leq \sum_{m' \neq 1} \sum_{\mathcal{C}} \Pr(\mathcal{C}) \Pr[(x^n(m'), Y^n) \in A_\epsilon^{(n)}(P_{XY})|m=1, \mathcal{C}] \\ &= (2^{nR} - 1) \Pr[(X^n(2), Y^n) \in A_\epsilon^{(n)}(P_{XY})|m=1] \\ &\doteq 2^{nR} 2^{-nI(P_X, W)}.\end{aligned}\tag{28}$$

The last inequality is from the fact that $(X^n(2), Y^n)$ is i.i.d. with $[P_X \cdot P_Y]$ and the property $[P_X \cdot P_Y]^n(A_\epsilon^{(n)}(P_{XY})) \doteq 2^{-nI(X;Y)}$, which we already proved for joint AEP. As long as $R < I(P_X, W)$, we can guarantee that this probability goes to 0. Thus, this completes the proof for $\Pr(\text{error}) \rightarrow 0$ for random coding.

Final Touch:

We have just proved that averaged over the random codebooks and over the possible messages, the probability of error can be made to decay to 0 as long as $R < I(P_X, W)$. To prove the coding theorem, we just need to a few more small steps.

- First, we will of course choose P_X to maximize $I(P_X, W)$, so as to guarantee $R < C := \max_{P_X} I(P_X, W)$ is achievable.
- Second, pick a sequence of (n, R) codes with $R < C$ such that $P_e^{(n)} \rightarrow 0$.
- For such a sequence of (n, R) codes with

$$\frac{1}{2^{nR}} \sum_{m=1}^{2^{nR}} P_{e,m} < \epsilon, \quad (29)$$

we claim that there exists at least $2^{nR}/2$ (a half of codewords) such that $P_{e,m} < 2\epsilon$ for all m . Otherwise, the codebook can never satisfy (29). Now we simply construct a new code with such $2^{nR}/2$ codewords. The new code has the maximum error probability goes to 0 as $n \rightarrow \infty$. Moreover, the loss of rate by reducing the codewords by half is ignorable when n is large enough since the new rate R' is $R' = \frac{1}{n} \log(2^{nR}/2) = \frac{1}{n} \log 2^{n(R-1/n)} = R - \frac{1}{n}$.

Remarks:

- 1) There are obviously many loose parts in this proof: a sub-optimal decoder, random coding as well as repeated usage of union bounds. But since we already proved the converse and showed that the achievability and converse meet at the common boundary $R = C$, such looseness in the proof does not matter.
- 2) The notion of i.i.d. entries with a certain distribution P_X^* in the codebook is quite mysterious. Our construction says that the codewords should all be chosen from the optimal input distribution P_X^* , or at least have an empirical distribution like that. It is not clear at this point whether this is necessary. The answer is in some sense blurred by the magical random coding argument. Clearly, randomized input is NOT a fundamental way to achieve reliable communication, but only a proof technique. But if the input distribution is a key concept, we must be able to see this from another way, without random coding.
- 3) Sphere packing picture
- 4) Clever use of union bound is necessary: if we do not divide the error types into $\mathcal{E}_1$ and $\mathcal{E}_2$ but just apply the union bound for pairwise error probability, we have

$$\begin{aligned} \Pr(\text{error} | m = 1) &= \Pr[\exists m' \neq 1 \text{ s.t. } x^n(m') \text{ appears more likely than } x^n(1)] \\ &\leq (2^{nR} - 1) \Pr[X^n(2) \text{ appears more likely than } X^n(1)]. \end{aligned} \quad (30)$$

It turns out that with this way we cannot prove the achievability for $R < C$. The reason is as follows. If $X^n(1)$ behaves in a atypical way, there are many other incorrect codewords that appear more likely than $X^n(1)$, and that makes this bound too loose to prove the achievability. The smart part of the proof we presented before is to separate the error event into two parts before applying the union bound for each incorrect codewords.